

Test of Mathematics for University Admission
Mock Set B Paper 1
 Applications of Mathematical Knowledge

Name: _____

Date sat: _____

Score: _____ / 20

[illegible]

Question 1

Find the exact value of $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7}$.

A $\frac{1}{16}$

B $\frac{1}{8}$

C $\frac{1}{4}$

D $\frac{\sqrt{7}}{8}$

E $-\frac{1}{8}$

F $\frac{1}{2}$

Question 2

Evaluate

$$\int_{-1}^5 |x^3 - 6x^2 + 12x - 9| dx.$$

A -6

B 6

C 24

D 36

E 42

F 48

Question 3

A polynomial p of degree 4 satisfies $p(k) = 2^k$ for $k = 0, 1, 2, 3, 4$. Find $p(5)$.

A 16

B 30

C 31

D 32

E 33

F 64

Question 4

Let a , b and c be the three real roots of $x^3 - 3x + 1 = 0$. Find the value of

$$(a^2 - 2)(b^2 - 2)(c^2 - 2).$$

- A -3
- B -1
- C 0
- D 1
- E 3
- F 9

Question 5

Positive real numbers x and y satisfy $x + y = 1$. Find the minimum value of

$$\left(x + \frac{1}{x}\right)^2 + \left(y + \frac{1}{y}\right)^2.$$

- A 8
- B $\frac{25}{4}$
- C $\frac{25}{2}$
- D 13
- E 15
- F 25

Question 6

Let M and m be the maximum and minimum values of

$$(\sin x + 2 \cos x)(2 \sin x + \cos x)$$

over all real x . Find Mm .

- A $-\frac{9}{4}$
- B $-\frac{5}{2}$
- C -1
- D 0
- E $\frac{9}{4}$
- F $\frac{81}{4}$

Question 7

Find the value of

$$\sum_{n=1}^{\infty} \frac{n+2}{n(n+1)2^n}.$$

- A $\frac{1}{2}$
- B $\frac{3}{4}$
- C 1
- D $\frac{5}{4}$
- E $\frac{3}{2}$
- F 2

Question 8

Find the sum of all solutions of

$$\sin x \cos x = \sin x + \cos x - 1$$

in the range $0^\circ \leq x \leq 360^\circ$.

- A 90°
- B 180°
- C 450°
- D 540°
- E 720°
- F 900°

Question 9

A regular heptagon has side length 1. Its diagonals take exactly two distinct lengths, $p < q$. Find the value of $\frac{1}{p} + \frac{1}{q}$.

- A $\frac{1}{2}$
- B $\frac{\sqrt{2}}{2}$
- C 1
- D $\frac{\sqrt{7}}{2}$
- E $\frac{3}{2}$

F 2**Question 10**

The roots of $x^2 - 2x - 1 = 0$ are α and β . Find the value of

$$\frac{1}{\alpha^5} + \frac{1}{\beta^5}.$$

A -82 **B** -41 **C** $-\frac{1}{82}$ **D** $\frac{1}{82}$ **E** 41 **F** 82 **Question 11**

There are exactly two straight lines that are tangent to both of the curves

$$y = x^2 \quad \text{and} \quad y = -x^2 + 6x - 10.$$

Find the product of the gradients of these two lines.

A -4 **B** -2 **C** -1 **D** 1 **E** 2 **F** 4 **Question 12**

Find the complete set of real values of x for which the infinite series

$$1 + (x - 1) + (x - 1)^2 + (x - 1)^3 + \dots$$

converges and has sum equal to $\frac{1}{2 - x}$.

A all real x with $x \neq 2$ **B** $x < 2$ **C** $0 < x < 2$ **D** $0 \leq x < 2$

E $1 < x < 2$

F $x > 0$

Question 13

How many chords of the circle $x^2 + y^2 = 25$ have length 6 and pass through the point $(3, 1)$?

A 0

B 1

C 2

D 3

E 4

F infinitely many

Question 14

A sequence is defined by $a_1 = 3$ and

$$a_{n+1} = \frac{a_n - 1}{a_n + 1} \quad (n \geq 1).$$

Find the value of the product $a_1 a_2 a_3 \cdots a_{2026}$.

A -2

B $-\frac{1}{2}$

C 1

D $\frac{3}{2}$

E 3

F $\frac{9}{4}$

Question 15

Find the complete set of real values of k for which the equation

$$4^x + 4^{-x} = k(2^x + 2^{-x}) - 4$$

has exactly two real solutions in x .

A $k \geq 2\sqrt{2}$

B $k > 2\sqrt{2}$

C $k \geq 3$

D $k > 3$

E $|k| > 3$

F $k > 2$

Question 16

The curve $y = x^4 - 4x^3 + ax^2$, where a is a real constant, has exactly one stationary point. Find the complete set of possible values of a .

A $a > \frac{9}{2}$

B $a \geq \frac{9}{2}$

C $a < \frac{9}{2}$

D $0 < a < \frac{9}{2}$

E $a > 0$

F $a \geq 0$

Question 17

Evaluate

$$\int_{-2}^2 (x+2)\sqrt{4-x^2} dx.$$

A 0

B 8

C 2π

D 4π

E 8π

F 16

Question 18

The points $A(1, 2)$ and $B(5, 2)$ are fixed, and C is a point such that $\angle ACB = 45^\circ$. Find the maximum possible area of triangle ABC .

A $2 + 2\sqrt{2}$

B $4\sqrt{2}$

C $4 + 2\sqrt{2}$

D $4 + 4\sqrt{2}$

E 8

F $8\sqrt{2}$

Question 19

A function f satisfies

$$\int_0^2 f(2x) \, dx = 3, \quad \int_1^3 f(x+1) \, dx = 5, \quad \int_0^1 f(3-3x) \, dx = 2.$$

Find the value of $\int_3^4 f(x) \, dx$.

- A -1
- B 0
- C 1
- D 2
- E 5
- F 6
- G cannot be determined from the information given

Question 20

Let $f(x) = x^3 + x$, and let f^{-1} denote the inverse function of f . Find the value of

$$\int_0^{10} f^{-1}(x) \, dx.$$

- A 6
- B 10
- C 14
- D 16
- E 20
- F 34

END OF PAPER